



Course: Analog and Digital Communication
 Program: BS(Electrical Engineering)
 Duration: 60 Minutes
 Paper Date: 21st February, 2017
 Section: EE
 Exam: Midterm-1

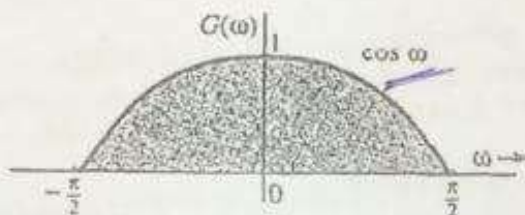
Course Code: EE323
 Semester: Spring 2017
 Total Marks: 30
 Weight: 15%
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Instruction/Notes:

Question # 1 [6+4 = 10 marks]

CLO 1

(a) Find the inverse Fourier transform of the following waveform using definition.



(b) The spectrum $G(\omega)$ is shifted in frequency by $\pm \omega_0$, plot $G(\omega - \omega_0)$ and $G(\omega + \omega_0)$ and find the Inverse Fourier transform of the new spectrum by using some property.

Question # 2 [10 marks]

CLO 2

For the following Gaussian pulse (widely used in communication modeling), verify Parseval's theorem by first finding the energy of the pulse by using $x(t)$.

$$x(t) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{t^2}{2\sigma^2}}$$

Hint: Use the fact that $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$

$$\frac{\omega \sigma^2}{2}$$

Question # 3 [6+4 = 10 marks]

CLO 3

Consider the following sinusoidal message $m(t)$ and carrier $c(t)$ signals,

$$m(t) = A_m \cos \omega_m t \text{ and } c(t) = A_c \cos \omega_c t, \text{ where } \omega_c \gg \omega_m$$

- Write down the expression for DSB-SC modulated signal, $g_{DSB-SC}(t)$. Also plot the spectra of the modulated signal $g_{DSB-SC}(t)$. Identify USB and LSB of the spectrum, $G_{DSB-SC}(\omega)$. For the message signal in hand, name the type of modulation.
- $g_{DSB-SC}(t)$ is demodulated using a coherent detection. Draw a labeled block diagram of the coherent detection with proper assigning of voltages before and after each block.